Signed Thue–Morse Prefix Sums and the Fabius Function

Sharp convergence rates, lattice deconvolution, and exact grid drift

A quantitative supplement to the formal development

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Abstract

Let $\varepsilon_m = (-1)^{\mathrm{wt}_2(m)}$ be the signed Thue–Morse sequence and let $\sigma_{k+1}(m) = \sum_{j=0}^{m} \sigma_k(j)$ be its inclusive iterated prefix sums. The formal development in `Analysis/FabiusFunction` proves that a shifted and normalized order-$(n+1)$ prefix row is exactly the coefficient row of a finite product of discrete uniform laws, and that its associated histogram converges pointwise to Rvachev's compactly supported function up. Sampling the same row at $\lfloor 2^n x \rfloor$ consequently converges to the bounded Fabius function $F(x)$. The existing argument is deliberately qualitative and supplies no rate.

This report makes the convergence quantitative. At the natural cell centers, the normalized prefix coefficients admit a uniform deconvolution expansion

$$H_n(\lambda) = \mathrm{up}(\lambda) - \frac{3n+4}{72\,4^n}\,\mathrm{up}''(\lambda) + \left(\frac{15n+16}{43200\,16^n} + \frac{(3n+4)^2}{10368\,16^n}\right)\mathrm{up}^{(4)}(\lambda) + \mathcal{O}\left(\frac{n^3}{64^n}\right).$$

It follows that the sharp uniform center-grid error is $\frac{1}{3}n4^{-n}(1+o(1))$; the same rate holds after piecewise-linear interpolation. The piecewise-constant histogram is less accurate because of cell quantization: $\|\mathcal{W}_n - \mathrm{up}\|_\infty = 2^{-n} + \mathcal{O}(n4^{-n})$. Most strikingly, the literal dyadic-floor approximation

$$A_n(x) = \frac{\sigma_{n+1}(\lfloor 2^n x \rfloor)}{2^{\binom{n}{2}}}$$

has a still larger, entirely geometric first-order error. Uniformly on $[0,1]$,

$$A_n(x) = F(x) + \frac{n+2-2\{2^n x\}}{2^{n+1}}F'(x) + \mathcal{O}\left(\frac{n^2}{4^n}\right),$$

and therefore

$$\left\|\frac{2^{n+1}}{n}(A_n - F) - F'\right\|_\infty \longrightarrow 0, \qquad \frac{2^n}{n}\|A_n - F\|_\infty \longrightarrow 1.$$

The factor n comes from the exact inward displacement of the finite support, not from slow weak convergence. At $x = 1/2$ the phenomenon has the closed form $A_n(1/2) - 1/2 = (n+2)2^{-n} - 2^{-\binom{n}{2}}$.

The bridge identities used as input are formalized in Lean. The Fourier asymptotic expansion, its sharp constants, and the accelerated finite-difference constructions developed here are new analytic deductions and are not claimed to have been formalized in the repository.

Contents

1 Scope, conventions, and the answer in one table

The source article [4] isolates the connection between iterated prefix sums and up, corrects two index/normalization defects in a naive formulation, and proves pointwise convergence. It also says explicitly that its monotonicity-and-continuity argument gives no uniform rate. The purpose of this supplement is to determine that rate, including its leading constant, and to explain why several superficially similar readings of the same integer row have different orders of accuracy.

We use the source article's normalization throughout:

- F is the bounded distribution function, equal to 0 on $(-\infty, 0]$ and to 1 on $[1, \infty)$;
- up is the even compactly supported density on $[-1, 1]$, with $\mathrm{up}(x-1) = F(x)$ for $0 \le x \le 1$;
- the Fourier transform is $\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) \mathrm{e}^{-2\pi \mathrm{i} x\xi}\, \mathrm{d}x$.

The sharp derivative theorem from the source gives

$$\|F^{(r)}\|_\infty = \|\,\mathrm{up}^{(r)}\,\|_\infty = 2^{\binom{r+1}{2}}. \tag{1}$$

In particular $\|F'\|_\infty = \|\,\mathrm{up}'\,\|_\infty = 2$ and $\|\,\mathrm{up}''\,\|_\infty = 8$.

The three natural rates. Put $h_n = 2^{-n}$. The same normalized prefix row produces three different errors.

Reading of the row	Sharp uniform error	Dominant mechanism
Natural center values, or their linear interpolant	$\frac{1}{3} n h_n^2 (1 + o(1))$	Deconvolution of the omitted mean-zero Bernoulli tail; its variance is of order $n h_n^2$.
Piecewise-constant histogram $\mathcal{W}_n$	$h_n + \mathcal{O}(n h_n^2)$	Horizontal quantization by half a cell, amplified by the optimal slope 2.
Dyadic-floor Fabius sample A_n	$n h_n (1 + o(1))$	The finite support is shorter than $[-1, 1]$ by $(n+2)h_n/2$ at each end, producing an inward drift.

Thus the pointwise floor formula is not the most accurate reconstruction of the function from the prefix row. It is worse than the histogram by a factor asymptotic to n, while a centered linear reconstruction is better than the histogram by a factor asymptotic to $3 \cdot 2^n / n$.

1.1 What is formalized and what is new here

The exact prefix convolution law, Prouhet cancellation, zero runs, generating-series identity, coefficient bridge, corrected cell centers, and pointwise limit are developed in `ThueMorsePrefix.lean` and `ThueMorseApproximation.lean`; the finite atomic and step approximants are developed in `EarlyApproximants.lean`, `EarlyMeasureBridge.lean`, `StepMeasureBridge.lean`, and `StepApproximationLimit.lean` [5]. These are the algebraic and measure-theoretic inputs used below.

The new ingredient is a quantitative inversion of the finite cosine product on its natural Nyquist interval. It yields a complete asymptotic expansion in even derivatives of up. The first two nonzero terms are enough to determine all sharp rates stated in the abstract. No part of that new expansion is represented here as already machine-checked.

2 The exact discrete bridge

2.1 Signed Thue–Morse prefixes

Let

$$\varepsilon_m = (-1)^{\mathrm{wt}_2(m)}, \qquad m \geq 0, \tag{2}$$

where $\mathrm{wt}_2(m)$ is the number of ones in the binary expansion of m. Define inclusive iterated prefix sums by

$$\sigma_0(m) = \varepsilon_m, \qquad \sigma_{k+1}(m) = \sum_{j=0}^{m} \sigma_k(j). \tag{3}$$

The word *inclusive* matters: one has

$$\sigma_{k+1}(m+1) - \sigma_{k+1}(m) = \sigma_k(m+1), \tag{4}$$

with the lower-order value at the new index.

Repeated summation is convolution with the discrete truncated-power kernel. For $k \geq 1$,

$$\sigma_k(m) = \sum_{r=0}^{m} \binom{m-r+k-1}{k-1} \varepsilon_r. \tag{5}$$

The signed generating series is

$$\sum_{m\geq 0} \varepsilon_m z^m = \prod_{j\geq 0}(1 - z^{2^j}), \qquad \sum_{m\geq 0} \sigma_k(m) z^m = \frac{\prod_{j\geq 0}(1 - z^{2^j})}{(1-z)^k}. \tag{6}$$

All identities in this paragraph are formal power-series identities over $\mathbb{Z}$.

2.2 The finite polynomial and the one-order shift

For $n \geq 0$ put

$$P_n(z) = \prod_{i=1}^{n}(1 + z + \cdots + z^{2^i - 1}), \qquad g_n = \deg P_n = 2^{n+1} - n - 2. \tag{7}$$

The cancellation of the first $n+1$ factors in (6) gives the exact coefficient identity

$$[z^m]P_n(z) = \sigma_{n+1}(m), \qquad 0 \leq m < 2^{n+1}. \tag{8}$$

This is the source of the order shift: the polynomial indexed by n corresponds to the $(n+1)$st, not the nth, prefix row.

Every factor of P_n is palindromic, hence

$$[z^m]P_n = [z^{g_n - m}]P_n, \qquad P_n(1) = 2^{\binom{n+1}{2}}. \tag{9}$$

The coefficients are nonnegative integers. Thus, after normalization, they form a symmetric probability mass function.

2.3 Natural centers, atomic law, and histogram

Set

$$h_n = 2^{-n}, \qquad \lambda_{n,m} = \frac{2m - g_n}{2^{n+1}} = h_n\left(m - \frac{g_n}{2}\right). \tag{10}$$

The points $\lambda_{n,m}$ lie on a translate of the lattice $h_n\mathbb{Z}$. Define

$$p_{n,m} = 2^{-\binom{n+1}{2}}[z^m]P_n, \qquad \mu_n = \sum_{m\in\mathbb{Z}} p_{n,m}\,\delta_{\lambda_{n,m}}, \tag{11}$$

where coefficients outside $0 \le m \le g_n$ are understood to be zero. Then μ_n is a probability measure. Its mass density relative to the lattice spacing is

$$H_n(\lambda_{n,m}) = \frac{p_{n,m}}{h_n} = \frac{[z^m]P_n}{2^{\binom{n}{2}}}. \tag{12}$$

By (8), this is exactly

$$H_n(\lambda_{n,m}) = \frac{\sigma_{n+1}(m)}{2^{\binom{n}{2}}}, \qquad 0 \le m < 2^{n+1}. \tag{13}$$

Let $I_{n,m}$ be the cell of width h_n centered at $\lambda_{n,m}$. Spreading each atom of μ_n uniformly over its cell gives the step density

$$\mathcal{W}_n(x) = \sum_m H_n(\lambda_{n,m})\mathbf{1}_{I_{n,m}}(x), \tag{14}$$

with half weight from each adjacent cell at a shared endpoint. In the interior of a cell, $\mathcal{W}_n$ is simply the corresponding normalized prefix coefficient. The formal repository proves

$$\mathcal{W}_n(x) \longrightarrow \mathrm{up}(x) \tag{15}$$

pointwise. The rest of this report quantifies (15) and the moving-center sample derived from it.

3 The hidden Bernoulli tail

3.1 Rademacher factorization

Let $R_{\ell,r}$ be independent Rademacher variables, each taking ± 1 with probability $1/2$. The binary factorization of every discrete uniform variable in (7) gives an equivalent representation of (11):

$$X_n = \sum_{\ell=1}^{n}\sum_{r=1}^{\ell} 2^{-\ell-1}R_{\ell,r}, \qquad \mathcal{L}(X_n) = \mu_n. \tag{16}$$

Consequently

$$\widehat{\mu_n}(\xi) = \prod_{\ell=1}^{n}\left(\cos\frac{\pi\xi}{2^\ell}\right)^{\ell}. \tag{17}$$

The complete series

$$Y = \sum_{\ell=1}^{\infty}\sum_{r=1}^{\ell} 2^{-\ell-1}R_{\ell,r} \tag{18}$$

converges absolutely, and its density is up. Equivalently,

$$\widehat{\mathrm{up}}(\xi) = \prod_{\ell=1}^{\infty}\left(\cos\frac{\pi\xi}{2^\ell}\right)^{\ell}. \tag{19}$$

This cosine representation is equivalent to the more familiar infinite sinc product.

Write

$$T_n = Y - X_n = \sum_{\ell > n} \sum_{r=1}^{\ell} 2^{-\ell-1} R_{\ell,r}. \tag{20}$$

Then X_n and T_n are independent and

$$\mathrm{up}(x)\,\mathrm{d}x = \mu_n * \mathcal{L}(T_n). \tag{21}$$

Thus μ_n is obtained by deleting a small, symmetric smoothing tail from the target law. The center-grid asymptotics are an exact lattice version of deconvolving that tail.

3.2 Exact support radius and variance

Proposition 3.1 (Geometry and second moment of the omitted tail)**.** *The tail T_n has mean zero, support contained in $[-r_n, r_n]$, and variance v_n, where*

$$r_n = \sum_{\ell > n} \ell 2^{-\ell-1} = \frac{n+2}{2^{n+1}}, \tag{22}$$

$$v_n = \sum_{\ell > n} \ell 2^{-2\ell-2} = \frac{3n+4}{36\,4^n}. \tag{23}$$

Both support endpoints occur.

Proof. The mean vanishes by symmetry. The maximal possible absolute value is obtained by taking all signs equal, so it is the first displayed sum. The geometric-tail identity

$$\sum_{\ell=n+1}^{\infty} \ell q^\ell = q^{n+1} \frac{(n+1) - nq}{(1-q)^2}$$

with $q = 1/2$ gives (22). Independence and $\mathrm{Var}(R_{\ell,r}) = 1$ give the variance sum, and the same identity with $q = 1/4$, followed by the extra factor $1/4$, gives (23). □

The support of μ_n is therefore

$$\operatorname{supp} \mu_n = [-1 + r_n, 1 - r_n] \cap \big(h_n(\mathbb{Z} - g_n/2)\big), \tag{24}$$

because $g_n/2^{n+1} = 1 - r_n$. Two scales should be distinguished:

$$r_n \sim \frac{n}{2} h_n, \qquad \sqrt{v_n} \sim \sqrt{\frac{n}{12}}\, h_n. \tag{25}$$

The support radius is a worst-case all-signs quantity, while the standard deviation is the typical scale. The floor-indexed approximation will inherit the former; the correctly centered deconvolution error will inherit the variance itself.

3.3 Why the leading center correction is second order

If a smooth density f is convolved with a small mean-zero law of variance v, then formally

$$f * \nu = f + \frac{v}{2} f'' + \cdots .$$

Solving backward gives $f - \frac{v}{2} f'' + \cdots$. Since (21) smooths the atomic row by the tail law, one should expect

$$H_n(\lambda) - \mathrm{up}(\lambda) \approx -\frac{v_n}{2} \mathrm{up}''(\lambda) = -\frac{3n+4}{72\,4^n} \mathrm{up}''(\lambda). \tag{26}$$

The sign is that of deconvolution: in a convex region the unsmoothed profile lies below the smoothed one. The next sections make (26) uniform and rigorous and compute the fourth-derivative term.

4 Fourier inversion on the moving lattice

The main technical point is that one must invert the finite atomic law on exactly one period of its lattice Fourier series. Ordinary Fourier inversion of a density cannot be applied directly to μ_n.

4.1 A shifted-lattice inversion formula

Lemma 4.1 (Nyquist inversion for a shifted lattice)**.** *Let $\mu = \sum_{k\in\mathbb{Z}} p_k \delta_{a+hk}$ be a finite measure on a translate of $h\mathbb{Z}$, and let $\widehat{\mu}(\xi) = \sum_k p_k \mathrm{e}^{-2\pi\mathrm{i}(a+hk)\xi}$. Then, for every $m \in \mathbb{Z}$,*

$$\frac{p_m}{h} = \int_{-1/(2h)}^{1/(2h)} \widehat{\mu}(\xi)\mathrm{e}^{2\pi\mathrm{i}(a+hm)\xi}\,\mathrm{d}\xi. \tag{27}$$

Proof. Insert the finite sum defining $\widehat{\mu}$ and integrate term by term. The kth term is

$$p_k \int_{-1/(2h)}^{1/(2h)} \mathrm{e}^{2\pi\mathrm{i}h(m-k)\xi}\,\mathrm{d}\xi,$$

which is p_m/h when $k = m$ and zero otherwise. The shift a cancels from the phase. □

For μ_n, the interval in (27) is

$$B_n = [-2^{n-1}, 2^{n-1}]. \tag{28}$$

Combining (12) and (27) gives

$$H_n(\lambda_{n,m}) = \int_{B_n} \widehat{\mu_n}(\xi)\mathrm{e}^{2\pi\mathrm{i}\lambda_{n,m}\xi}\,\mathrm{d}\xi. \tag{29}$$

4.2 The exact deconvolution multiplier

On B_n, all factors in the omitted cosine tail are positive: for $\ell > n$, $|\pi\xi/2^\ell| \le \pi/4$. Hence one may divide (19) by (17) and write

$$\widehat{\mu_n}(\xi) = \widehat{\mathrm{up}}(\xi)R_n(\xi), \qquad R_n(\xi) = \prod_{\ell>n}\left(\sec\frac{\pi\xi}{2^\ell}\right)^{\ell}, \qquad \xi \in B_n. \tag{30}$$

The multiplier R_n is the reciprocal characteristic function of the omitted tail on the frequency range where lattice inversion needs it.

There is a useful global bound. Put $t = \pi\xi/2^n$, so $|t| \le \pi/2$. Reindexing $\ell = n + r$ and using Viète's product $\prod_{r\ge1}\cos(t/2^r) = \sin t/t$ gives

$$R_n(\xi) = \left(\frac{t}{\sin t}\right)^n \prod_{r=1}^{\infty}\sec(t/2^r)^r. \tag{31}$$

The second product is bounded uniformly for $|t| \le \pi/2$, while $t/\sin t \le \pi/2$. Thus there is an absolute C_* such that

$$1 \le R_n(\xi) \le C_*\left(\frac{\pi}{2}\right)^n, \qquad \xi \in B_n. \tag{32}$$

This exponential growth is harmless because $\widehat{\mathrm{up}}$ decays faster than every power. Indeed, compact support and smoothness give, for each $M \ge 1$,

$$|\widehat{\mathrm{up}}(\xi)| \le \frac{\|\,\mathrm{up}^{(M)}\,\|_1}{(2\pi|\xi|)^M} \le \frac{2^{1+\binom{M+1}{2}}}{(2\pi|\xi|)^M} \qquad (\xi \ne 0), \tag{33}$$

where (1) was used in the second inequality.

5 Sharp center-grid asymptotics

5.1 Expansion of the reciprocal tail product

For $|u| < \pi/2$,

$$\log \sec u = \frac{u^2}{2} + \frac{u^4}{12} + \mathcal{O}(u^6). \tag{34}$$

The two geometric tails needed below are

$$\frac{1}{2}\sum_{\ell>n} \ell 4^{-\ell} = \frac{3n+4}{18\,4^n} =: \alpha_n, \tag{35}$$

$$\frac{1}{12}\sum_{\ell>n} \ell 16^{-\ell} = \frac{15n+16}{2700\,16^n} =: \beta_n. \tag{36}$$

Therefore, on a low-frequency range growing sufficiently slowly with n,

$$\log R_n(\xi) = \alpha_n(\pi\xi)^2 + \beta_n(\pi\xi)^4 + \mathcal{O}\Big(\frac{n}{64^n}|\xi|^6\Big), \tag{37}$$

and exponentiation gives

$$R_n(\xi) = 1 + \alpha_n(\pi\xi)^2 + \Big(\beta_n + \frac{\alpha_n^2}{2}\Big)(\pi\xi)^4 + \mathcal{O}\Big(\frac{n^3}{64^n}|\xi|^6\Big). \tag{38}$$

5.2 The uniform expansion

Theorem 5.1 (Two-term lattice deconvolution expansion). *Uniformly for all lattice centers $\lambda = \lambda_{n,m}$,*

$$\begin{aligned} H_n(\lambda) = {}& \mathrm{up}(\lambda) - \frac{3n+4}{72\,4^n}\,\mathrm{up}''(\lambda) \\ &+ \Big(\frac{15n+16}{43200\,16^n} + \frac{(3n+4)^2}{10368\,16^n}\Big)\,\mathrm{up}^{(4)}(\lambda) + \mathcal{O}\Big(\frac{n^3}{64^n}\Big). \end{aligned} \tag{39}$$

The implicit constant is absolute and effective. In particular,

$$H_n(\lambda) - \mathrm{up}(\lambda) = -\frac{3n+4}{72\,4^n}\,\mathrm{up}''(\lambda) + \mathcal{O}\Big(\frac{n^2}{16^n}\Big) \tag{40}$$

uniformly on the moving lattice.

Proof. Use (29) and (30) and split the integral at $|\xi| = 2^{n/4}$. On the inner interval, (38) is uniform. Since $\int_{\mathbb{R}} |\xi|^6 |\widehat{\mathrm{up}}(\xi)|\,\mathrm{d}\xi < \infty$, its integrated remainder is $\mathcal{O}(n^3 64^{-n})$ uniformly in the phase λ.

On the complementary part of B_n, combine (32) with (33). Taking, for example, $M = 32$ makes the resulting integral $o(64^{-n})$. The tails of the three polynomial Fourier integrals outside B_n are even smaller. We may therefore replace the truncated integrals of the displayed polynomial terms by integrals over all of $\mathbb{R}$.

With the chosen Fourier convention,

$$\int_{\mathbb{R}} (\pi\xi)^2 \widehat{\mathrm{up}}(\xi) \mathrm{e}^{2\pi \mathrm{i}\lambda\xi}\,\mathrm{d}\xi = -\frac{1}{4}\,\mathrm{up}''(\lambda), \qquad \int_{\mathbb{R}} (\pi\xi)^4 \widehat{\mathrm{up}}(\xi) \mathrm{e}^{2\pi \mathrm{i}\lambda\xi}\,\mathrm{d}\xi = \frac{1}{16}\,\mathrm{up}^{(4)}(\lambda).$$

Substituting (35) and (36) gives (39). Omitting the fourth-derivative term and using (1) gives (40). □

Remark 5.2 (Variance form). Because $\alpha_n = 2v_n$, the leading term is exactly

$$H_n = \mathrm{up} - \frac{v_n}{2}\,\mathrm{up}'' + \cdots, \tag{41}$$

which confirms the deconvolution heuristic (26).

5.3 Sharp norm constant

Corollary 5.3 (Sharp center-grid rate). *One has the uniform profile limit*

$$\sup_{m\in\mathbb{Z}}\left|\frac{4^n}{n}\big(H_n(\lambda_{n,m})-\mathrm{up}(\lambda_{n,m})\big)+\frac{1}{24}\,\mathrm{up}''(\lambda_{n,m})\right|\longrightarrow 0. \tag{42}$$

Consequently,

$$\lim_{n\to\infty}\frac{4^n}{n}\sup_{m\in\mathbb{Z}}|H_n(\lambda_{n,m})-\mathrm{up}(\lambda_{n,m})|=\frac{1}{3}. \tag{43}$$

Proof. The coefficient $(3n+4)/(72n)$ tends to $1/24$, while the remainder in (40), after multiplication by $4^n/n$, tends uniformly to zero. The lattices become dense, and up'' is continuous with exact supremum 8. This supremum is attained; for example, the attained-point form of the sharp derivative theorem gives $\mathrm{up}''(-3/4)=F''(1/4)=8$. Hence the norm limit is $8/24=1/3$. □

5.4 A continuous reconstruction with the same rate

Let L_n be the piecewise-linear interpolant through the points $(\lambda_{n,m},H_n(\lambda_{n,m}))$, with zero values continued outside the finite support. The ordinary interpolation error for a C^2 function is at most $h_n^2\|\,\mathrm{up}''\,\|_\infty/8=h_n^2$. This is smaller by a factor $1/n$ than the leading center bias.

Corollary 5.4 (Sharp continuous linear reconstruction). *Uniformly on* $\mathbb{R}$,

$$\frac{4^n}{n}(L_n-\mathrm{up})\longrightarrow-\frac{1}{24}\,\mathrm{up}'', \tag{44}$$

in the supremum norm. In particular,

$$\|L_n-\mathrm{up}\,\|_\infty=\frac{n}{3\,4^n}(1+o(1)). \tag{45}$$

Thus iterated prefix sums do not merely converge to up after a qualitative passage through histograms. Read at their actual centers and interpolated linearly, they give a continuous approximation with a fully determined, curvature-shaped leading error.

5.5 The all-orders pattern

The same proof is not tied to fourth order. Write

$$\log\sec z=\sum_{r\ge1}c_rz^{2r}\quad(c_1=1/2,\ c_2=1/12,\ c_3=1/45,\ldots) \tag{46}$$

and put

$$s_{r,n}=c_r\sum_{\ell>n}\ell 2^{-2r\ell}. \tag{47}$$

For a fixed M, expand

$$\exp\left(\sum_{r=1}^{M-1}s_{r,n}z^{2r}\right)=\sum_{j=0}^{M-1}b_{j,n}z^{2j}+\mathcal{O}(z^{2M}). \tag{48}$$

Then the lattice inversion argument gives

$$H_n(\lambda)=\sum_{j=0}^{M-1}\frac{(-1)^jb_{j,n}}{4^j}\,\mathrm{up}^{(2j)}(\lambda)+\mathcal{O}_M\big(n^M4^{-Mn}\big)\,, \tag{49}$$

uniformly on the center lattice. Each $b_{j,n}$ is an explicit polynomial in the geometric tails $s_{r,n}$. Thus the connection admits a full asymptotic differential expansion, not just a big-O convergence estimate.

6 The histogram: a sharp first-order quantization error

The center values are exceptionally accurate, but $\mathcal{W}_n$ holds each value constant across a cell. That reconstruction injects a larger first-order horizontal error.

For a point x in the interior of a cell, let $c_n(x)$ denote that cell's center and define the sawtooth phase

$$\rho_n(x) = \frac{c_n(x) - x}{h_n}, \qquad -\frac{1}{2} < \rho_n(x) < \frac{1}{2}. \tag{50}$$

Then (40) and Taylor's theorem give, uniformly away from the shared cell endpoints,

$$\mathcal{W}_n(x) - \operatorname{up}(x) = h_n \rho_n(x) \operatorname{up}'(x) + \mathcal{O}(nh_n^2). \tag{51}$$

At a shared endpoint, the half-weight convention averages the two adjacent center values and cancels the first-order odd displacement, leaving only $\mathcal{O}(nh_n^2)$ there.

Theorem 6.1 (Sharp histogram rate)**.** *The step approximants satisfy*

$$\|\mathcal{W}_n - \operatorname{up}\|_\infty = 2^{-n} + \mathcal{O}\Big(\frac{n}{4^n}\Big). \tag{52}$$

Equivalently,

$$\lim_{n\to\infty} 2^n \|\mathcal{W}_n - \operatorname{up}\|_\infty = 1. \tag{53}$$

Proof. For an interior point, combine (40) with the optimal Lipschitz bound $\|\operatorname{up}'\|_\infty = 2$:

$$|\mathcal{W}_n(x) - \operatorname{up}(x)| \le \mathcal{O}(nh_n^2) + 2|c_n(x) - x| \le h_n + \mathcal{O}(nh_n^2).$$

At a shared endpoint the averaged value obeys the same bound. There remains only the thin uncovered strip between the support of $\mathcal{W}_n$ and an endpoint of $[-1,1]$. Its width is at most $r_n + h_n/2 = \mathcal{O}(nh_n)$. Since up and all of its derivatives vanish at ± 1, Taylor's theorem with the sharp bounded third derivative gives

$$|\operatorname{up}(\pm 1 \mp s)| \le \frac{\|\operatorname{up}^{(3)}\|_\infty}{6} s^3 = \mathcal{O}(n^3 h_n^3) = o(nh_n^2) \qquad (0 \le s \le r_n + h_n/2).$$

Outside $[-1,1]$ both functions vanish. This completes the upper estimate globally.

For the lower estimate, choose cells whose centers tend to $-1/2$, where $\operatorname{up}'(-1/2) = 2$, and approach one edge of each chosen cell from its interior. The center error is $\mathcal{O}(nh_n^2)$, while the change of up across half a cell is $h_n + \mathcal{O}(h_n^2)$. Taking the supremum gives $\|\mathcal{W}_n - \operatorname{up}\|_\infty \ge h_n - \mathcal{O}(nh_n^2)$. ☐

Remark 6.2. The pointwise proof in the source article could not reveal (52): weak convergence plus unimodality controls values through fixed comparison points but does not track a moving point at distance $h_n/2$ from a cell center. The rate is a local geometric phenomenon, and the sharp constant comes from the sharp derivative bound.

7 The floor-prefix approximation to the bounded Fabius function

7.1 Definition and exact moving-center displacement

The source theorem obtains F from the prefix row by evaluating its up approximation at half the argument. Define

$$A_n(x) = \mathcal{J}_n(x/2) = \frac{\sigma_{n+1}(\lfloor 2^n x \rfloor)}{2^{\binom{n}{2}}}, \qquad 0 \le x \le 1. \tag{54}$$

Let

$$\theta_n(x) = \{2^n x\} = 2^n x - \lfloor 2^n x \rfloor. \tag{55}$$

The normalized prefix value in (54) is not located at $x-1$. Its actual center is

$$\begin{aligned}\lambda_n(x) &= \lambda_{n,\lfloor 2^n x\rfloor} \\ &= x - 1 + \delta_n(x), \end{aligned}\tag{56}$$

$$\delta_n(x) = \frac{n+2-2\theta_n(x)}{2^{n+1}}. \tag{57}$$

This identity, not merely the fact that $\lambda_n(x) \to x-1$, determines the dominant error. It also gives

$$\frac{n}{2^{n+1}} < \delta_n(x) \le \frac{n+2}{2^{n+1}}. \tag{58}$$

In terms of the omitted-tail support radius (22),

$$\delta_n(x) = r_n - \theta_n(x) h_n. \tag{59}$$

Thus the index $\lfloor 2^n x \rfloor$ compensates for the ordinary fractional cell position but not for the shortening of the finite support. The residual is roughly $r_n \sim n h_n/2$, which is much larger than either the cell width error of the histogram or the variance-scale center bias.

7.2 First- and second-order expansions

Theorem 7.1 (Uniform floor-prefix expansion)**.** *Uniformly for $x \in [0,1]$,*

$$A_n(x) = F(x) + \delta_n(x) F'(x) + \mathcal{O}\left(\frac{n^2}{4^n}\right). \tag{60}$$

More precisely,

$$\begin{aligned} A_n(x) = F(x) &+ \delta_n(x) F'(x) \\ &+ \left(\frac{\delta_n(x)^2}{2} - \frac{3n+4}{72\,4^n}\right) F''(x) + \mathcal{O}\left(\frac{n^3}{8^n}\right). \end{aligned}\tag{61}$$

All remainder constants are independent of x and n.

Proof. By the exact coefficient identification, $A_n(x) = H_n(\lambda_n(x))$. Apply (40) at $\lambda_n(x) = x - 1 + \delta_n(x)$, use $\mathrm{up}(x-1) = F(x)$ and the same identity for derivatives, and Taylor-expand around $x-1$. Since $\delta_n = \mathcal{O}(n2^{-n})$, the first omitted Taylor term in (60) is $\mathcal{O}(n^2 4^{-n})$.

For (61), retain the quadratic Taylor term and the leading center-bias term. The remainders are $\mathcal{O}(\delta_n^3)$, $\mathcal{O}(n4^{-n}\delta_n)$, and $\mathcal{O}(n^2 16^{-n})$, all absorbed by $\mathcal{O}(n^3 8^{-n})$. □

Corollary 7.2 (Uniform profile and sharp norm rate)**.** *The entire rescaled error profile converges uniformly:*

$$\left\| \frac{2^{n+1}}{n}(A_n - F) - F' \right\|_\infty \longrightarrow 0. \tag{62}$$

Consequently,

$$\lim_{n\to\infty} \frac{2^n}{n} \|A_n - F\|_\infty = 1. \tag{63}$$

For each fixed x,

$$\lim_{n\to\infty} \frac{2^{n+1}}{n}\big(A_n(x) - F(x)\big) = F'(x). \tag{64}$$

Proof. By (57),

$$\frac{2^{n+1}}{n}\delta_n(x) = 1 + \frac{2-2\theta_n(x)}{n} \longrightarrow 1$$

uniformly in x. Multiplying the remainder in (60) by $2^{n+1}/n$ gives $\mathcal{O}(n2^{-n})$. This proves (62). Taking norms and using $\|F'\|_\infty = 2$ gives (63). □

7.3 The equivalent statement for the up sample

The source notation uses
$$\mathcal{J}_n(u) = \frac{\sigma_{n+1}(\lfloor 2^{n+1}u \rfloor)}{2^{\binom{n}{2}}}. \tag{65}$$
For $0 \le u < 1$ put $\vartheta_n(u) = \{2^{n+1}u\}$ and
$$d_n(u) = \frac{n + 2 - 2\vartheta_n(u)}{2^{n+1}}. \tag{66}$$
Then
$$\mathcal{J}_n(u) = \mathrm{up}(2u - 1) + d_n(u)\,\mathrm{up}'(2u - 1) + \mathcal{O}\left(\frac{n^2}{4^n}\right). \tag{67}$$
At $u = 1$, the prefix coefficient identity is outside its range and the exact zero-run value is $-2^{-\binom{n}{2}}$; since $\mathrm{up}(1) = \mathrm{up}'(1) = 0$, this exceptional endpoint is much smaller than the uniform principal scale and does not change the norm asymptotic.

8 An exact midpoint identity

The sharp constant in (63) can be seen without asymptotic analysis at the single point where F' attains its maximum.

Theorem 8.1 (Exact floor-prefix error at $x = 1/2$)**.** *For every $n \ge 2$,*
$$A_n\left(\frac{1}{2}\right) = \frac{1}{2} + \frac{n+2}{2^n} - 2^{-\binom{n}{2}}. \tag{68}$$
Equivalently,
$$\frac{2^n}{n}\left(A_n\left(\frac{1}{2}\right) - \frac{1}{2}\right) = 1 + \frac{2}{n} - \frac{2^{n-\binom{n}{2}}}{n}. \tag{69}$$
Proof. Put $M = 2^{n-1}$ and write $Q = P_{n-1} = \sum_{r=0}^{d} q_r z^r$, where $d = g_{n-1} = 2^n - n - 1$, extending q_r by zero outside $0 \le r \le d$. Since $M < 2^n$, multiplication by the last factor of P_n gives
$$[z^M]P_n = \sum_{r=0}^{M} q_r =: C. \tag{70}$$
The polynomial Q is palindromic and has total mass
$$T = Q(1) = 2^{\binom{n}{2}}. \tag{71}$$
By symmetry,
$$2C - T = \sum_{r=d-M}^{M} q_r. \tag{72}$$
The band in (72) has $n + 2$ coefficients. To evaluate them, write $Q = P_{n-2}(1 + \cdots + z^{2^{n-1}-1})$. Put $T_0 = P_{n-2}(1) = 2^{\binom{n-1}{2}}$. The n interior coefficients of the band are all T_0, because the convolution window contains all coefficients of P_{n-2}. Each of the two boundary coefficients is $T_0 - 1$: its window omits only the monic leading coefficient of P_{n-2}, and palindromicity gives the same value at the opposite boundary. Hence
$$2C - T = (n + 2)T_0 - 2. \tag{73}$$
Since $T/T_0 = 2^{n-1}$,
$$\frac{C}{T} = \frac{1}{2} + \frac{n+2}{2^n} - \frac{1}{T}.$$
Finally, (8) and (54) identify C/T with $A_n(1/2)$, proving (68). ∎

Table 1: Exact midpoint error and its sharp-rate normalization.

n	$A_n(1/2) - 1/2$	$2^n\big(A_n(1/2) - 1/2\big)/n$
2	0.5	1.000000000000
3	0.5	1.333333333333
4	0.359375	1.437500000000
5	0.2177734375	1.393750000000
6	0.124969482421875	1.333007812500
8	0.0390624962747097	1.249999880791
10	0.0117187499999716	1.199999999997
12	0.0034179687500000	1.166666666667
16	0.0002746582031250	1.125000000000
20	0.00002098083496094	1.100000000000

The exact identity also explains an otherwise puzzling numerical observation: after a few levels, the error is almost indistinguishable from the tangent displacement $2\delta_n(1/2) = (n+2)2^{-n}$. The remaining defect is the super-exponentially small term $2^{-\binom{n}{2}}$.

8.1 Correct centering can be exact at the same point

The large error in (68) is not an unavoidable defect of the prefix row. It is caused by reading the wrong index as the point $-1/2$ on the up axis. When n is even, $-1/2$ itself is a lattice center, with index

$$m_n^* = 2^{n-1} - \frac{n+2}{2}. \tag{74}$$

For this index, $m_n^* = (g_{n-1} - 1)/2$. Since P_{n-1} is palindromic of odd degree, exactly half of its total coefficient mass lies at indices at most m_n^*. The last-factor prefix identity therefore gives

$$H_n(-1/2) = \frac{1}{2} = \mathrm{up}(-1/2) \qquad (n \text{ even}). \tag{75}$$

The floor rule instead uses $M = 2^{n-1}$, which is $(n+2)/2$ whole cells to the right of m_n^*. This is the discrete form of the support drift.

9 Bias removal and accelerated prefix reconstructions

The expansion (39) can be used constructively. It tells us how to remove the leading deconvolution bias using only neighboring prefix coefficients.

For a function on the center lattice, define the centered second difference

$$\Delta_h^2 H(\lambda) = \frac{H(\lambda + h_n) - 2H(\lambda) + H(\lambda - h_n)}{h_n^2}. \tag{76}$$

Put

$$c_n = \frac{3n+4}{72} h_n^2 \tag{77}$$

and define the corrected grid

$$H_n^{[1]}(\lambda) = H_n(\lambda) + c_n \Delta_h^2 H_n(\lambda). \tag{78}$$

Every value in (78) is a rational linear combination of three normalized iterated-prefix values.

Theorem 9.1 (One-step deconvolution acceleration)**.** *Uniformly on the center lattice,*

$$H_n^{[1]}(\lambda) - \mathrm{up}(\lambda) = -\frac{n^2}{1152\,16^n}\,\mathrm{up}^{(4)}(\lambda) + o\left(\frac{n^2}{16^n}\right). \tag{79}$$

Consequently,

$$\lim_{n\to\infty} \frac{16^n}{n^2} \sup_{\lambda \in h_n(\mathbb{Z}-g_n/2)} |H_n^{[1]}(\lambda) - \mathrm{up}(\lambda)| = \frac{8}{9}. \tag{80}$$

Proof. For a smooth function,

$$\Delta_h^2 f = f'' + \frac{h_n^2}{12} f^{(4)} + \mathcal{O}(h_n^4). \tag{81}$$

Write (39) as $H_n = \mathrm{up} - c_n\,\mathrm{up}'' + d_n\,\mathrm{up}^{(4)} + \mathcal{O}(n^3 h_n^6)$, where

$$d_n = \left(\frac{15n+16}{43200} + \frac{(3n+4)^2}{10368}\right) h_n^4.$$

Substitute this into (78) and use (81). If the uniform remainder is denoted by R_n, then $\|\Delta_h^2 R_n\|_\infty \le 4h_n^{-2}\|R_n\|_\infty$; hence $c_n \Delta_h^2 R_n = \mathcal{O}(n^4 h_n^6) = o(n^2 h_n^4)$. Thus the remainder remains negligible at the claimed scale. The second-derivative terms cancel, and the coefficient of $\mathrm{up}^{(4)}$ is

$$d_n - c_n^2 + \frac{c_n h_n^2}{12} = -\frac{n^2}{1152} h_n^4 + \mathcal{O}(n h_n^4).$$

This proves (79). By (1), $\|\,\mathrm{up}^{(4)}\,\|_\infty = 2^{10} = 1024$, so the sharp norm constant is $1024/1152 = 8/9$. □

Higher corrections follow by replacing the reciprocal-tail multiplier with successively longer polynomials in the lattice Laplacian. The all-orders expansion (49) supplies the coefficients recursively. To preserve the accelerated rate in a continuous reconstruction, one should use interpolation of sufficiently high order: piecewise-linear interpolation has an $\mathcal{O}(h_n^2)$ geometric error and would mask the $\mathcal{O}(n^2 h_n^4)$ corrected lattice error, whereas local cubic interpolation has $\mathcal{O}(h_n^4)$ error and retains the latter as the dominant term.

10 Conceptual synthesis

10.1 One row, three coordinate choices

The normalized integer row

$$m \longmapsto \frac{\sigma_{n+1}(m)}{2^{\binom{n}{2}}} \tag{82}$$

contains highly accurate samples, but the meaning of m must be specified.

1. The intrinsic coordinate is the center $\lambda_{n,m} = h_n(m - g_n/2)$. At that coordinate the row differs from up by a curvature term of order nh_n^2.
2. Holding a center value constant across its width-h_n cell changes the coordinate by at most $h_n/2$. The target has maximal slope 2, so the histogram error is h_n.
3. Replacing the intrinsic coordinate by the naive dyadic coordinate implicit in $m = \lfloor 2^n x \rfloor$ overlooks the support deficit r_n. It changes the coordinate by roughly $nh_n/2$, so the Fabius error is roughly $nh_n F'(x)/2$.

The three rates are therefore compatible, not contradictory. They measure distinct reconstructions of the same discrete data.

10.2 Variance versus support radius

The center expansion sees the omitted tail only through its even cumulants. Its mean is zero, so the first nonzero effect is its variance $v_n \asymp nh_n^2$. The floor-coordinate drift, by contrast, sees the extreme support loss $r_n \asymp nh_n$. This is why a centered analytic argument is exponentially more accurate than a support-aligned but miscentered index rule.

This distinction is common in lattice approximation. Weak convergence controls averages, and moment matching controls smooth centered observables, but an endpoint-derived coordinate can retain a worst-case support defect long after the bulk distribution has become extremely close.

10.3 Why the n is structural

The degree

$$g_n = 2^{n+1} - n - 2 \tag{83}$$

is shorter than the naive value 2^{n+1} by exactly $n + 2$. After scaling by 2^{n+1} and centering, half of this deficit appears at each endpoint. Hence the factor n in the floor rate is already encoded in the elementary degree formula for P_n. No delicate property of F is needed to predict its presence; the analytic work is needed to turn the horizontal deficit into the sharp vertical profile F' and to control the smaller terms.

10.4 A practical recommendation

For numerical reconstruction from a computed prefix row:

- do not use $m/2^n$ as the physical coordinate;
- attach each value to $\lambda_{n,m}$ from (10);
- use at least linear interpolation rather than a histogram;
- if another factor of roughly $4^n/n$ is valuable, apply the three-point correction (78) and use cubic interpolation.

The coefficients remain exact rationals until the final interpolation or evaluation stage.

11 A Lean formalization roadmap

The new proof separates naturally into reusable lemmas. A formalization need not begin with the full asymptotic theorem.

1. **Shifted-lattice Fourier inversion.** Formalize theorem 4.1 for a finitely supported measure on $a + h\mathbb{Z}$. This is a finite-sum orthogonality calculation.
2. **Rademacher realization.** Refine the existing finite-measure bridge to the triangular Rademacher array (16); obtain (17) and the independent tail decomposition (20).
3. **Exact geometric tails.** Add the closed forms (22), (23), (35), and (36). These reduce to finite manipulations of geometric series.
4. **Uniform $\log\sec$ remainder.** Prove a sixth-order Taylor bound on $[-1/4, 1/4]$ and use it scale by scale. The low-frequency cutoff can be chosen as $2^{n/4}$ or replaced by any convenient dyadic integer bound.
5. **Rapid Fourier decay.** Package repeated integration by parts for compactly supported C^∞ functions and instantiate it with the already formalized sharp derivative bounds.

6. **Center expansion.** Combine the preceding lemmas with Fourier derivative identities. The first-order version (40) is substantially shorter than the fourth-order theorem and already proves all three basic rates.
7. **Exact drift.** Formalize (56) by integer-floor arithmetic. Then (60) is a direct Taylor estimate.
8. **Midpoint identity.** Prove theorem 8.1 independently of Fourier analysis from palindromicity and the central coefficient plateau. It is an especially useful regression test for index conventions.

The formal theorem statements should distinguish three domains: values at the center lattice, the step function on all reals, and the floor sample on $[0, 1]$. Combining them into one statement would obscure exactly the coordinate issue responsible for the different rates.

12 Conclusion

Iterated summation of the signed Thue–Morse sequence does more than produce a sequence that happens to resemble the Fabius function. After the correct one-order shift, its finite row is an exact lattice deconvolution of up. The omitted part is an explicit triangular Bernoulli tail with support radius $(n + 2)2^{-n-1}$ and variance $(3n + 4)/(36\,4^n)$. Those two exact numbers govern two different approximation regimes.

At the natural centers, the mean-zero tail leaves a variance-scale curvature bias, yielding the sharp rate $n/(3\,4^n)$. Holding the values constant across cells introduces the larger sharp rate 2^{-n}. Reading the same row at the naive dyadic floor leaves the full support deficit as a coordinate drift and yields the still larger sharp rate $n2^{-n}$. The latter has the uniform error profile F' and an exact midpoint formula.

The main conceptual correction is therefore simple but consequential: the integer index is not itself the continuum coordinate. Once the genuine center map is retained, the Thue–Morse prefix construction is far more accurate than its original pointwise formulation suggests, and its error admits a complete, explicitly computable differential expansion.

A Elementary geometric-tail formulas

For $|q| < 1$ and $n \geq 0$,

$$\sum_{\ell=n+1}^{\infty} \ell q^{\ell} = q\frac{\mathrm{d}}{\mathrm{d}q}\left(\frac{q^{n+1}}{1-q}\right) = q^{n+1}\frac{(n+1) - nq}{(1-q)^2}. \tag{84}$$

The specializations used in the text are

$$\sum_{\ell>n} \ell 2^{-\ell} = (n+2)2^{-n}, \tag{85}$$

$$\sum_{\ell>n} \ell 4^{-\ell} = \frac{3n+4}{9\,4^n}, \tag{86}$$

$$\sum_{\ell>n} \ell 16^{-\ell} = \frac{15n+16}{225\,16^n}, \tag{87}$$

$$\sum_{\ell>n} \ell 64^{-\ell} = \frac{63n+64}{3969\,64^n}. \tag{88}$$

These identities make every coefficient in the all-orders expansion elementary: no unevaluated infinite tail sum is necessary.

B A short exact-arithmetic algorithm

The coefficient row $[z^m]P_n$ can be built without polynomial multiplication. Multiplication by $1 + \cdots + z^{2^i-1}$ is a sliding-window sum. The following pseudocode maintains exact integers and costs $\mathcal{O}(2^n)$ arithmetic operations per level.

```
coeff := [1]
for i := 1,...,n:
    width := 2^i
    nextLength := length(coeff) + width - 1
    next := array(nextLength, 0)
    window := 0
    for m := 0,...,nextLength-1:
        if m < length(coeff):
            window := window + coeff[m]
        if m >= width:
            window := window - coeff[m-width]
        next[m] := window
    coeff := next

# Normalized prefix sample:
H_n(m) := coeff[m] / 2^(n(n-1)/2)
# Physical coordinate:
lambda_n(m) := (2m - (2^(n+1)-n-2)) / 2^(n+1)
```

The midpoint identity (68), symmetry $\texttt{coeff}[\texttt{m}] = \texttt{coeff}[\texttt{g}_\texttt{n} - \texttt{m}]$, the total mass $2^{\binom{n+1}{2}}$, and the central maximum $2^{\binom{n}{2}}$ provide independent exact checks of an implementation.

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